An open dataset on Magic The Gathering Cards for teaching biostatistics.

Zuliani and Lortie
in prep for Journal of Statistics and Data Science Education
(https://www.tandfonline.com/journals/ujse21)

fit

JSDSE has a specific article category called “Datasets and Stories”, designed for papers introducing interesting multivariate datasets for teaching statistics, including learning outcomes, potential pitfalls, and classroom applications. It also explicitly covers statistics education in the health sciences.

dataset

Zuliani, Mario; lortie, cj (2020). Magic The Gathering Data. figshare. Dataset. https://doi.org/10.6084/m9.figshare.12797474.v2 (https://doi.org/10.6084/m9.figshare.12797474.v2) (need to update with new file in email)

7,713 Card observations

763 Unique card names

3 MTG sets

540 Pack sampling units

4 Rarity classes

prices in 2021 and now 2026

Many teaching Qs (here are three)

1. Are rare cards really rare?

Yes. Of 7,713 cards sampled, 71.0% were common, 21.5% uncommon, 6.6% rare, and just 0.9% mythic.

2. Are MTG booster packs a good investment?

Usually not. Assuming a 4.99 purchase price, the average pack’s 2026 card value was 6.00, but the median was only 3.60. Just 33.6% of fully priced packs exceeded the purchase price, before resale fees.

3. What was the best investment five years later?

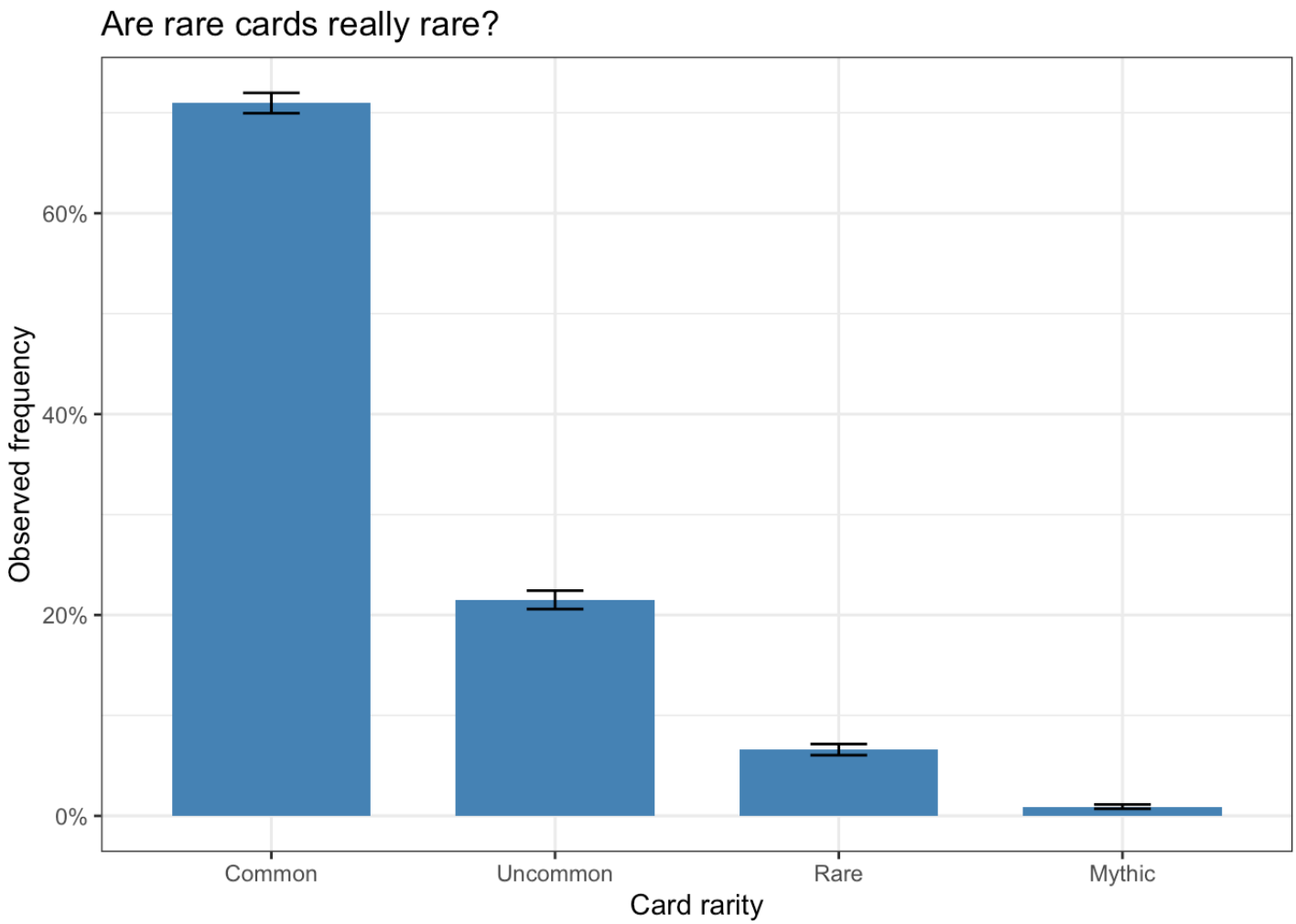
Based on the change from original prices to 2026 prices:
- Best set: Core 2021 (–9.2%).
- Best card type: Instant (+67.6%).
- Best combination: Zendikar Rising × Instant (+213.6%).

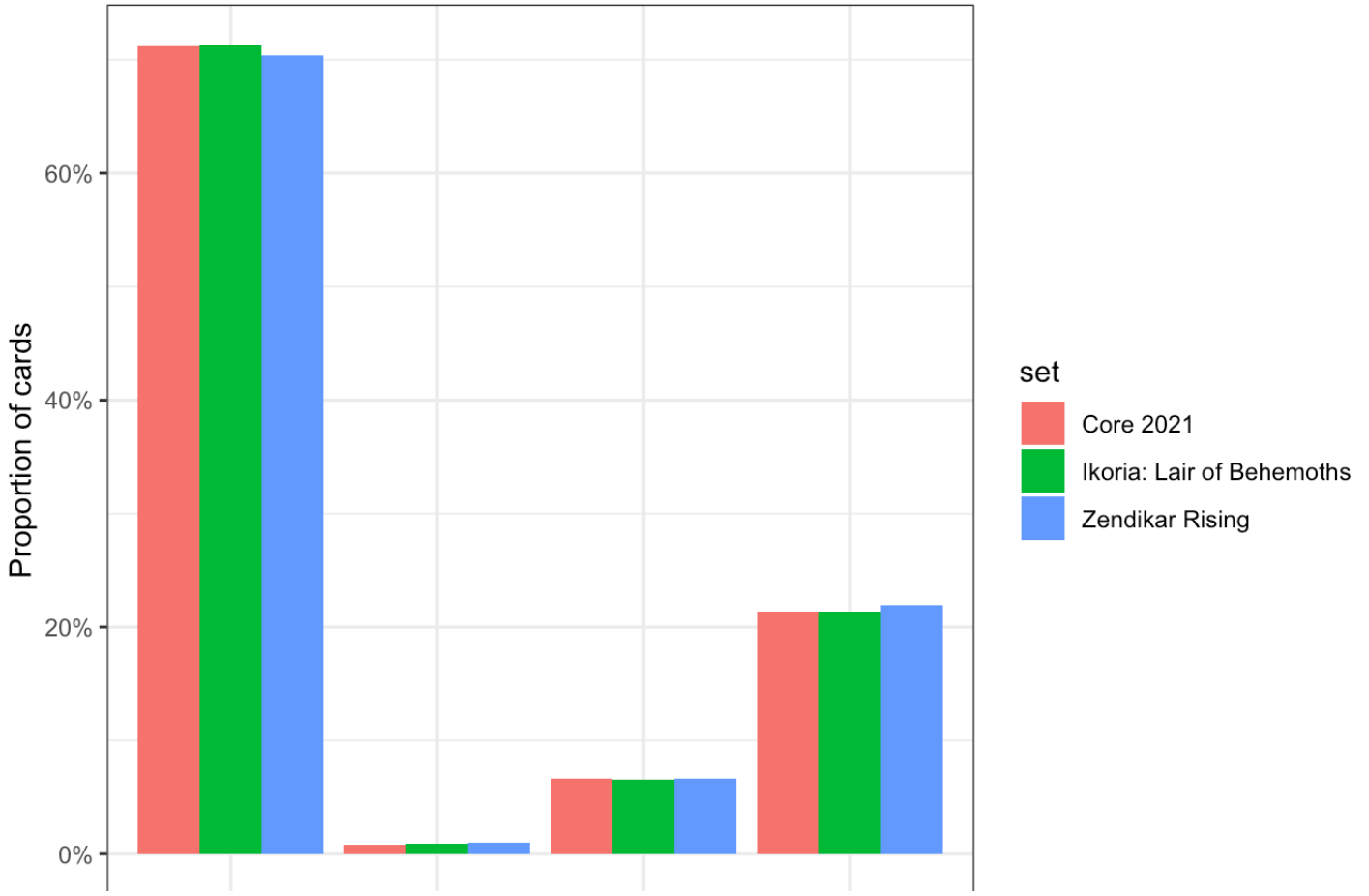
Note: The sets were released in 2020, so this represents approximately six years to 2026, although the exact date of the original prices is unknown.

Are rare cards really rare?

```
##
##  Chi-squared test for given probabilities
##
## data:  table(df$rarity)
## X-squared = 9391.2, df = 3, p-value < 2.2e-16
```

```
##
##  Pearson's Chi-squared test
##
## data:  table(df$set, df$rarity)
## X-squared = 1.1609, df = 6, p-value = 0.9788
```


Are rare cards really rare?
Observed frequency
60%
40%
20%
0%
Common
Uncommon
Rare
Mythic
Card rarity

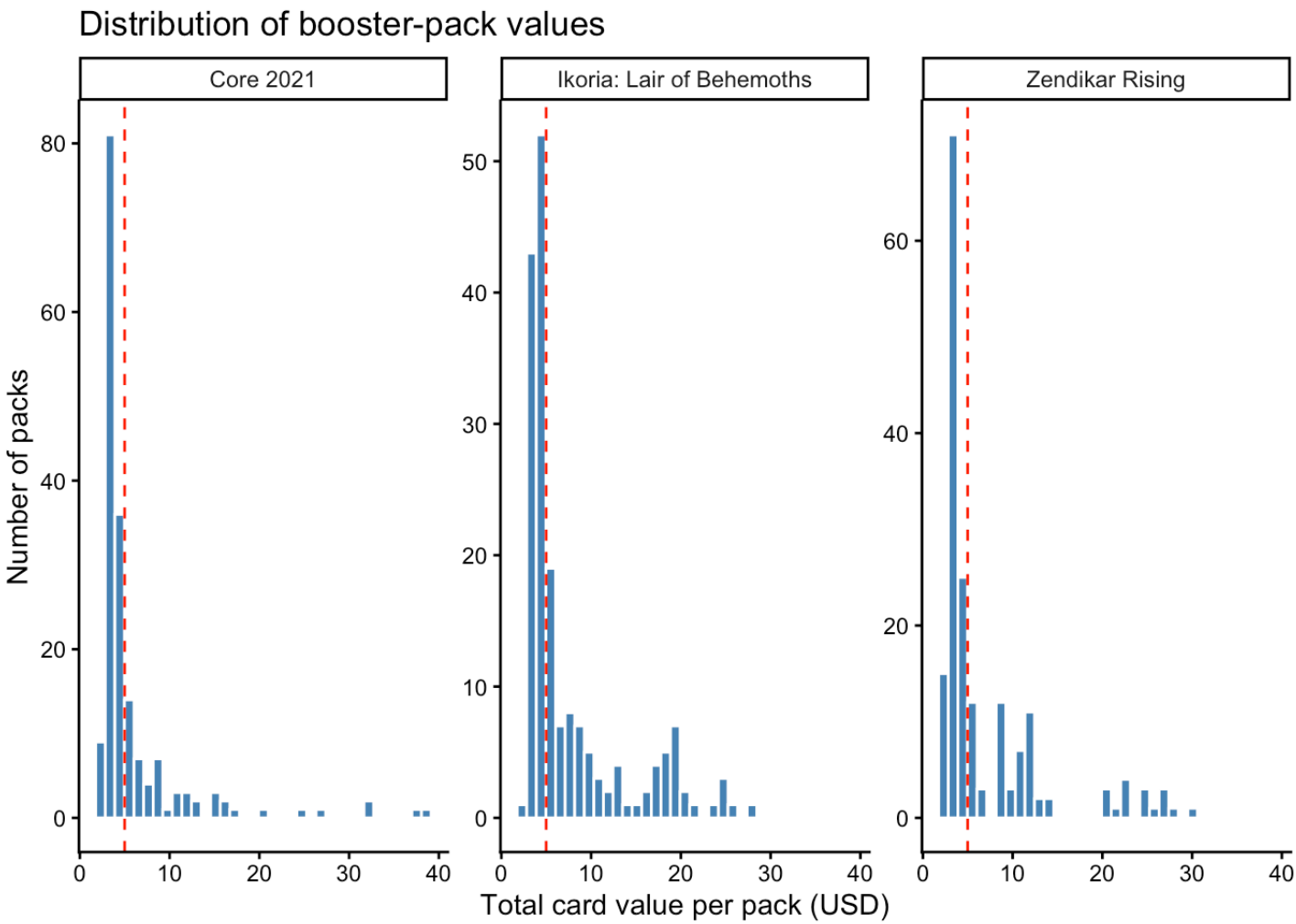

Proportion of cards
60%
40%
20%
0%
set
Core 2021
Ikoria: Lair of Behemoths
Zendikar Rising



Are MTG booster packs a good investment?

```
##
##  One Sample t-test
##
## data:  packs$pack_value
## t = 7.3187, df = 539, p-value = 9.146e-13
## alternative hypothesis: true mean is not equal to 4.99
## 95 percent confidence interval:
##  6.388432 7.414531
## sample estimates:
## mean of x
##  6.901481
```

```
##     2.5%    97.5%
## 1.408100 2.419539
```

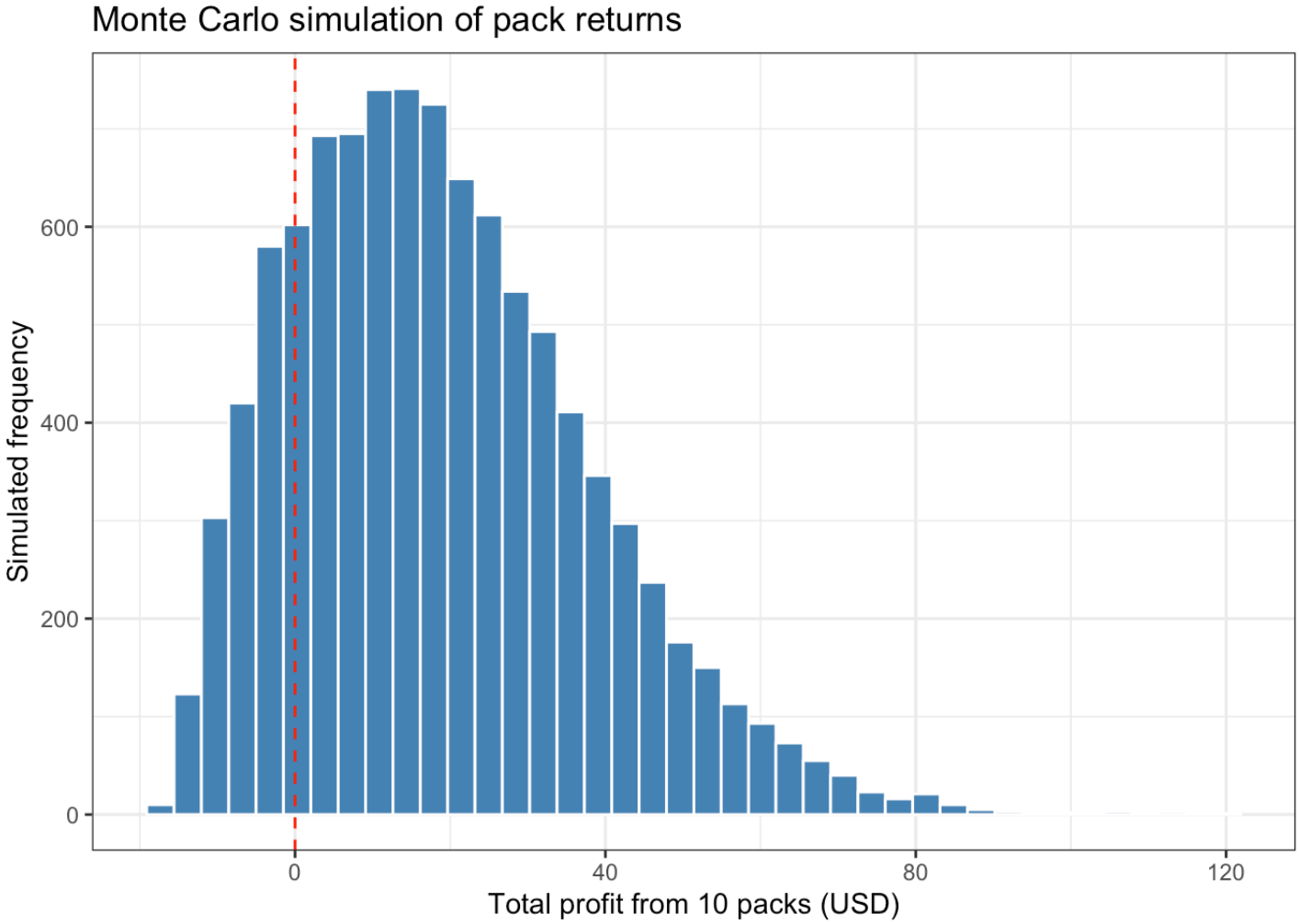



```
## # A tibble: 2 × 4
##   mythic_present mean_value median_value     n
##   <lgl>                <dbl>        <dbl> <int>
## 1 FALSE                 5.62          4     471
## 2 TRUE                 15.7          12.7    69
```

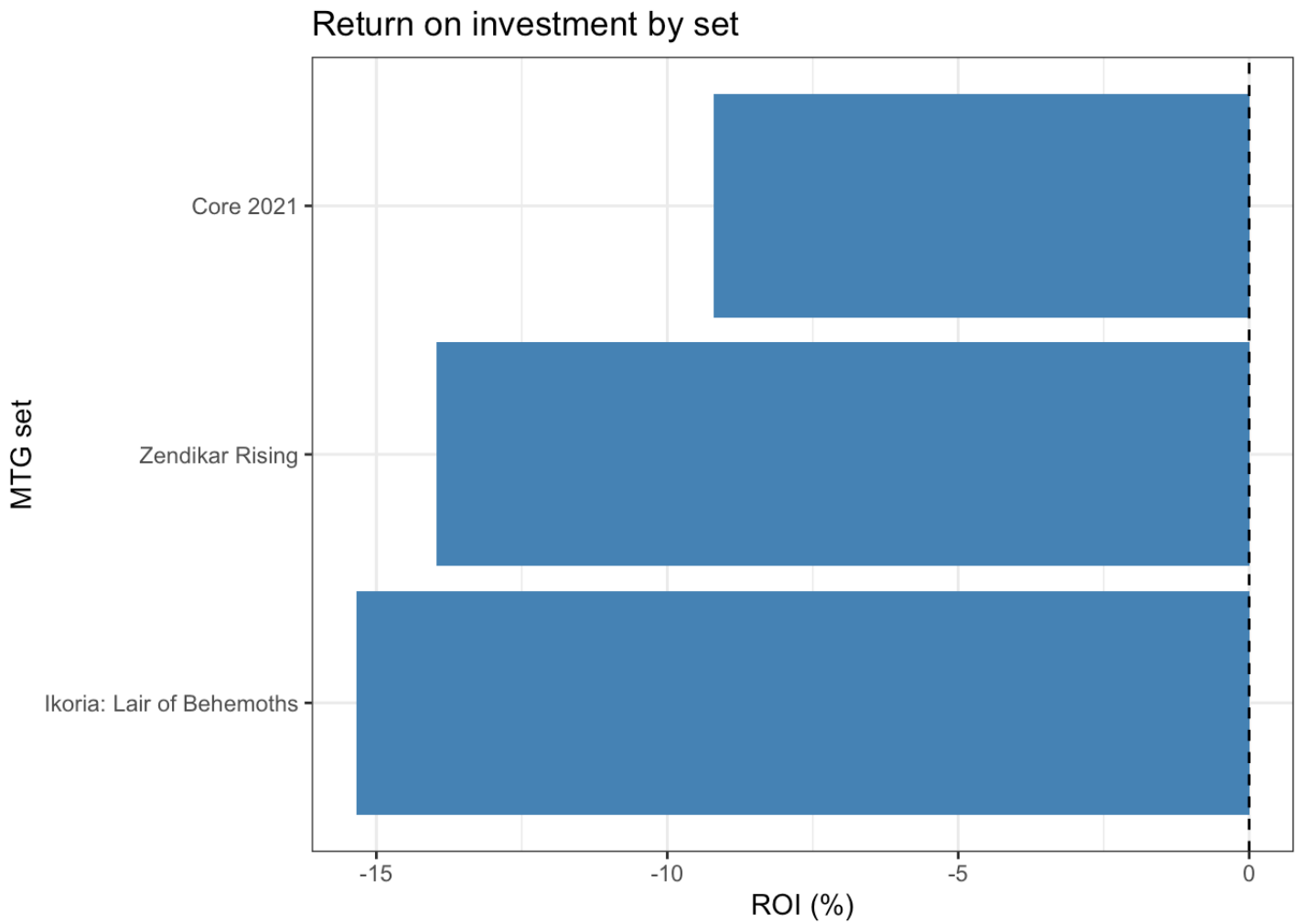
```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  pack_value by mythic_present
## W = 3399, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal t
o 0
```

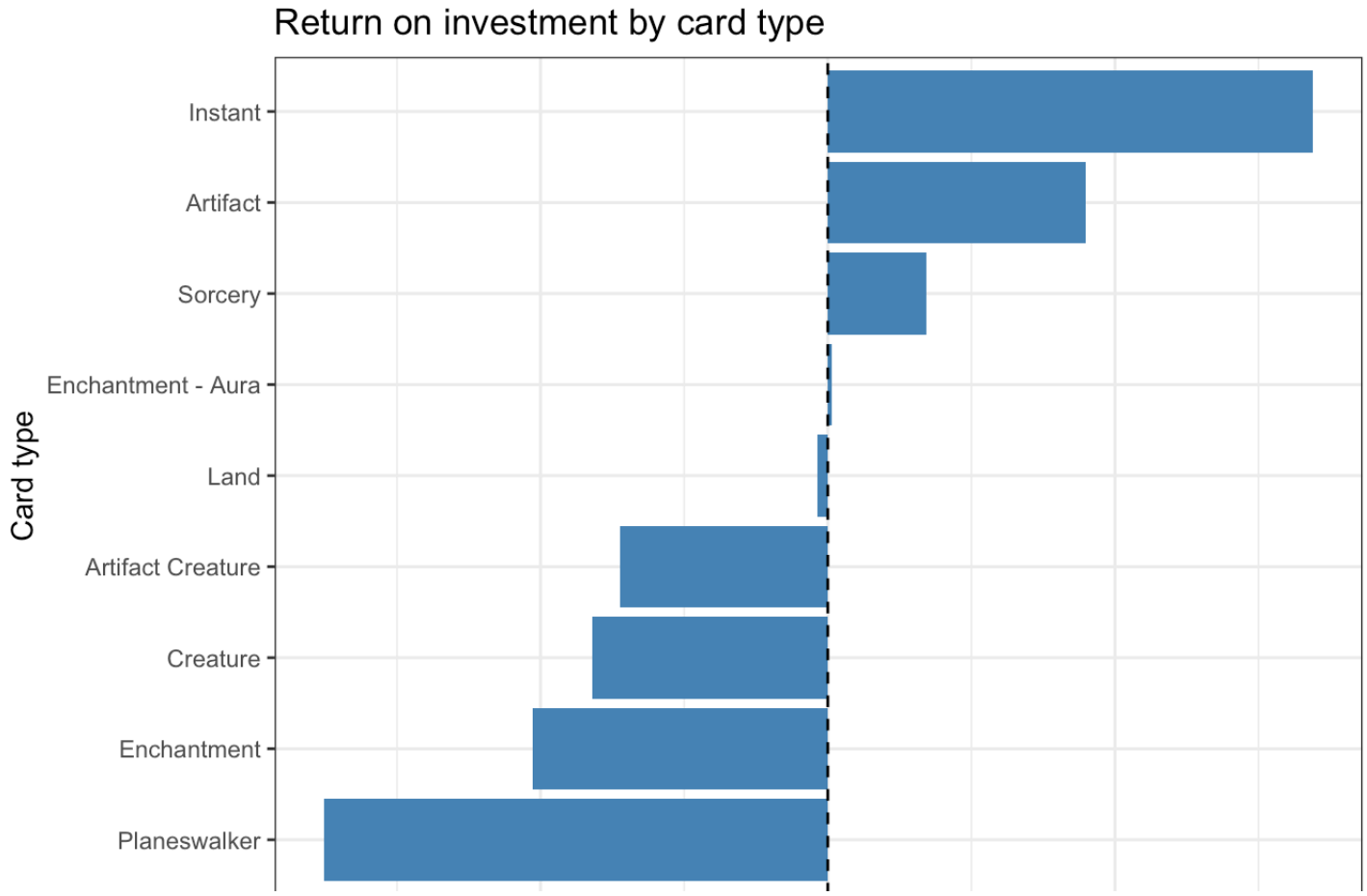
```
## Mean profit for 10 packs: 18.90025
```

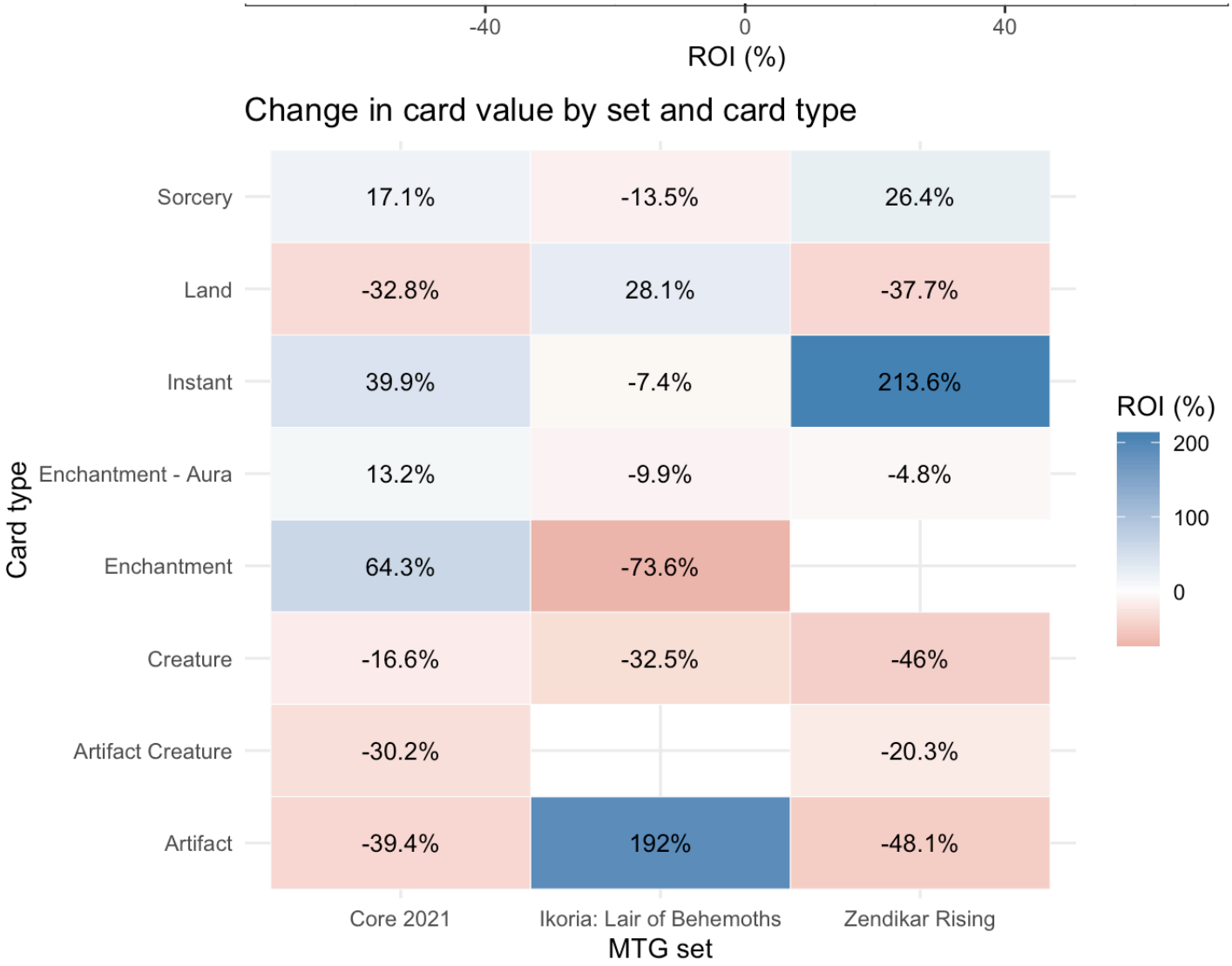
```
## Probability of profit: 0.8324
```



What was the best investment 5-years later?


Return on investment by set
MTG set
Core 2021
Zendikar Rising
Ikoria: Lair of Behemoths
-15
-10
-5
0
ROI (%)


Return on investment by card type
Card type
Instant
Artifact
Sorcery
Enchantment - Aura
Land
Artifact Creature
Creature
Enchantment
Planeswalker


-40
0
40
ROI (%)
Change in card value by set and card type
Sorcery
Land
Instant
Enchantment - Aura
Enchantment
Creature
Artifact Creature
Artifact
17.1%
-13.5%
26.4%
-32.8%
28.1%
-37.7%
39.9%
-7.4%
213.6%
13.2%
-9.9%
-4.8%
64.3%
-73.6%
-16.6%
-32.5%
-46%
-30.2%
-20.3%
-39.4%
192%
-48.1%
Card type
Core 2021
Ikoria: Lair of Behemoths
Zendikar Rising
MTG set
ROI (%)
200
100
0